



Society of Petroleum Engineers

**SPE-230001-MS**

## **An Innovative System to Revive Liquid-Loaded Vertical Gas Wells**

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/230001-MS](https://doi.org/10.2118/230001-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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### **Abstract**

Reviving liquid-loaded gas wells is critical to sustaining the production of hydrocarbon gas and condensate. As formation liquids, namely water and condensate, accumulate in a gas well, they can restrict flow and effectively "suffocate" the well, preventing valuable gas from reaching the surface.

This paper introduces a new, synergistic technique that builds upon the recent research advancements. The approach integrates two distinct, yet complementary technologies: a patented Flow Stabilizing Device (FSD) and a patented electric treatment system. The FSD facilitates the removal of liquids by locally modifying the gas phase velocity within the well, stabilizing the liquid film on the tubular walls, and promoting an ideal annular flow regime. To further enhance well performance, the FSD is combined with an electric treatment method. This involves applying an electric current at the wellhead, which propagates along the metallic casing to the reservoir. The current induces localized physical and electrochemical effects.

More details on the proposed synergistic approach, including the schematics and principal mechanisms, are provided. The electrical treatment operation takes up to 30 hours, which can permanently remove the near wellbore water/condensate blockage due to effective release and mobilization of all the water/condensate clogging the pore throats. The process also simultaneously treats two liquid-loaded gas wells by flowing the electric current between the two wells in the reservoir using formation water as the conductive medium. Additionally, partial electrolysis generates gas bubbles that help displace trapped liquids, while improved connectivity among water or condensate droplets facilitates their mobilization and removal. The FSD then prevents reverse liquid flow and enables accelerated transport of liquids to the surface, even in wells that are fully loaded with water/condensate. The combination of these two technologies thereby works synergistically to revive and prolong the life of gas wells. In addition, the paper also sheds some insights on successive steps to be followed to eventually move this synergistic technology towards practical applications.

The combined use of the FSD and electric treatment becomes a useful practical solution for restoring production in liquid-loaded gas wells. This integrated approach enhances gas well performance, reduces water cut, and extend the productive life of gas wells.

## Introduction

Reviving liquid loaded gas well is essential for continued production of gas and liquid condensate. When a gas well is loaded with formation liquids; namely, condensate and water, the well suffocates preventing the valuable hydrocarbon gas to flow out of the well. Most gas reservoirs are associated with underlying or surrounding aquifers (Ogolo et al., 2014), and as the gas production is declined, water encroachment and increasing water production typically occur over time. Elevated water production in gas wells can eventually lead to water loading, which may halt gas flow entirely (Roozshenas et al., 2021). Chemical stimulation of a water-blocked gas well can help mobilize trapped liquids, enhancing their flow and improving gas production by increasing the gas flow rate (Ahmadi et al., 2021).

As gas saturation decreases due to water blockage, the flowing bottomhole pressure (FBHP) can drop below the dewpoint pressure of the reservoir fluid, causing the well to cease production (Afidick et al., 1994). Once water begins to block the near-wellbore region, it tends to accumulate over time, worsening the severity of the issue (Engineer, 1985). Numerous case studies have been conducted to better understand gas and condensate flow within reservoirs and to evaluate the impact of liquid blockage in gas wells (Blom and Hagoort, 1998; Whitson et al., 1999; Pope et al., 2000; Kamath, 2007; Jamiolahmady et al., 2009; Fussell, 1973; Clark, 1985; Hinchman and Barree, 1985; Henderson et al., 1993; Fevang and Whitson, 1996; Mott, 1999).

The new technique presented here build on teachings from two prior works developed by the authors (Al-Juhani et al., 2024, and Shariff et al., 2021). The uniqueness of the technology presented in this paper describes when combining the two techniques we will be able to revive the gas well even if the gas well is completely liquid loaded with condensate and water. Such remedial intervention is achieved by employing the Flow Stabilizing Device (FSD). Once the liquid loading the well is reduced, the well gets produced by accelerating the liquids up the vertical production tubular in a perfectly annular flow regime. The device described in (Shariff et al., 2021) acts to locally modify the gas phase velocity in the well so that the liquid film formed at the walls is kept stable and reverse liquid flow is avoided. When combined with the electric treatment described by (Al-Juhani et al., 2024) it will further enhance reviving dead gas wells challenged by liquid blockage. The electrical treatment is initiated by a power supply emitting electric current. After current is applied to well head, the charge will propagate through metallic casing along the well until pay zone delivering to reservoir.

When electric current is applied, a mechanical destruction of the blocked pore space which results an increase in permeability. Also, the increase in pore throat radius due to local electric pulses will create a space for water to move and be released from pore spaces. The occurrence of partial electrolysis due to passing of electric current leads to gas bubbles formation from released liquids. Electric current would also improve the connectivity of released water droplets for easy mobilization and removal. The objectives of this paper are to improve and enhance the gas recovery, irreversibly enlarging the pore radius by removing the blockage which therefore increases the permeability, and finally to reduce water cut, as described by (Al-Juhani et al., 2024). This would revive and prolong the life of gas well for several years when combined with the device described by (Shariff et al., 2021).

In detail, the FSD and Electrical treatment technology will be explained, highlighting their individual functionalities and contributions. The aim of the work presented in this study is to produce and sustain a desired two-phase annular flow regime to transport liquid and gas upward inside vertical pipes of circular cross-section by incorporating one or several Flow Stabilizing Device (FSD) configurations in the pipe.

### Flow Stabilizing Device (FSD)

The FSD accelerates the flow in vertical pipes at critical locations where additional momentum is needed to sustain the upward liquid flow. The FSD configuration can be applied in production columns, and natural gas producing wells with liquid loading problems, to enhance liquid lifting in gas wells. Figure 1 shows a

schematic of the FSD. Thanks to its converging-diverging geometry, FSD increases the gas flow velocity, which increases momentum transfer to the liquid phase.

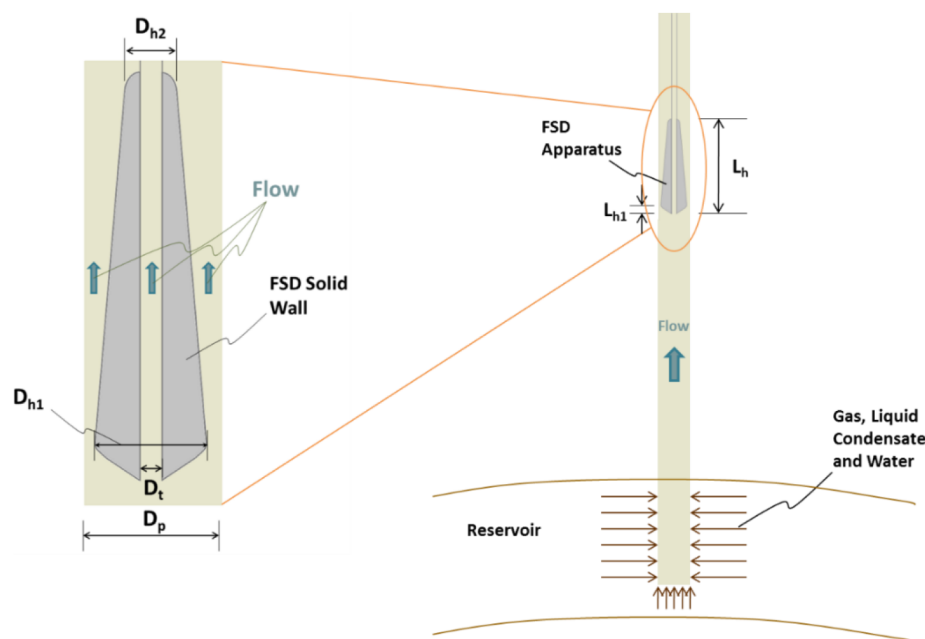


Figure 1—Schematic of the FSD in operation (Shariff et al., 2021)

The FSD is supported and held in position inside the production column by a supporting center tube. The FSD is fabricated from a rigid corrosion resistant material. The material can be a metal, sleeved metal or non-metallic. A pneumatic, hydraulic or electrical mechanism is used to adjust the vertical location of FSD and fix the position the FSD at the desired height in the production column.

In one realization of the FSD, the annular throat is adjustable via an upward/downward sliding mechanism to change  $D_{h1}$  and thus control momentum and liquid entrainment as required by the flow conditions. In another version of the FSD, the outer surface of the FSD is flexible allowing, with an internal mechanism, the adjustment of the throat gap to change  $D_{h1}$  and thus control momentum and liquid entrainment as required by the flow conditions. The FSD can be centered in the production column by any means such as centering mechanical device located on the FSD supporting tube or by means of ribs or centering winglets attached to some of the cone inserts forming the FSD.

The FSD can also be used in combination with a pressure control device to split the ascending flow inside the column into a gas rich flow through a central tube at the tip of the FSD with diameter  $D_t$  in Figure 2, and a gas+liquid annular flow around the FSD. The pressure control device can be any system capable of keeping a constant static pressure difference between the center tube and the annular flow section, at the top of the production column, or the well head. This type of FSD is named as the Hollow Flow Stabilizing Device or HFSD. The foreseen advantage of the HFSD is a way to optimize the production process by continuously adjusting the static pressure difference to maximize the total gas production rate.

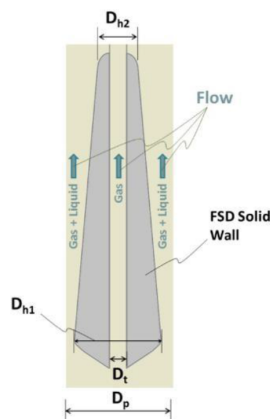


Figure 2—Schematics of the flow around the Hollow Flow Stabilizing Device operating inside a vertical gas well (Shariff et al., 2021)

In another form, the FSD is used in combination with a pressurizing system capable of injecting a gas or a liquid through the center tube into the production column in order to help stabilizing the liquid film at the wall of the production column. In this version, the FSD can be equipped with holes or slots distributed in any part of the cone inserts or above the cone inserts, in the center tube supporting them. This FSD type is called as the Injection Flow Stabilizing Device or IFSD. The foreseen advantage of the IFSD is the assisted flow stabilizing effect by means of pressurized gas injection (added gas lift effect resulting in enhanced liquid film stabilization at the column wall) or/and chemical additive injection (low dosage liquid film stabilization agent or viscosifier or surfactant). Figure 3 shows different orthographic views of the FSD configuration. Figure 4 shows the primary geometry and dimensions of the FSD.

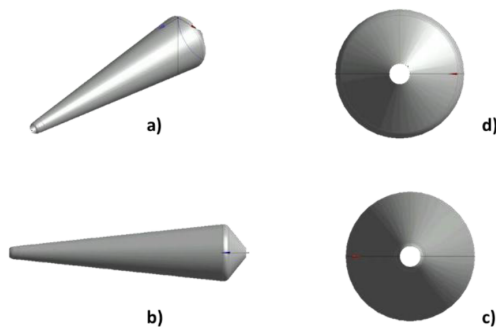


Figure 3—Orthographic views of FSD: (a) Perspective view; (b) Side view; (c) Top view; (d) Bottom view (Shariff et al., 2021)

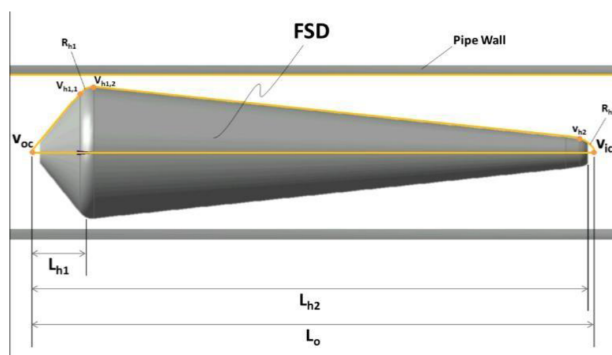


Figure 4—Schematic of the FSD geometry (Shariff et al., 2021)

## Electrical Treatment

The main objective when deploying the electrical treatment to a dead gas well is to mainly enhance the production rate by removing the water blockage. The proposed idea of electrical treatment operation requires power supply, a transformer to step down power by transferring from one circuit to another through mechanical induction, a power unit and a protection controlled automatic equipment for disconnection in case of safety hazard. Cables lines are necessary to connect all aforementioned equipment's together. When applying electric treatment operation, a high voltage power supply is needed as a source of energy of 200-500KVA. The power transformer is used to lower the voltage to the required value of 100-120 volts. The cable wires have the following characteristics to decrease the amount of electricity lost while travelling through cables:

- Flexible
- Containing of four thin copper wires with a cross section 120 square millimeters (total cross section of 480 square Single-core millimetres)
- Rubber or plastic coating
- Not exceed 500 meters length

There have been several studies that described the electrical treatment operation. Electrical current remediation to revive liquid-loaded gas wells due to fluid obstruction was studied by ([Al-Juhani et al., 2023](#)). Deployment of electrostimulation techniques in certain oil fields were presented by ([Batirbaey et al., 2000](#)). To improve recovery, lower water cut, reduce H<sub>2</sub>S production, and increase API gravity of oil ([Rehman and Meribout, 2012](#)), direct electrical current is applied in heavy-oil reservoirs. Electro-enhanced oil recovery (EOR) using direct current was also investigated by ([Wittle et al., 2007](#)).

A schematic view of the electrical treatment method when applied between two vertical wells (one as anode and one as cathode) to remove water blockage in gas wells is represented in [Figure 5](#).

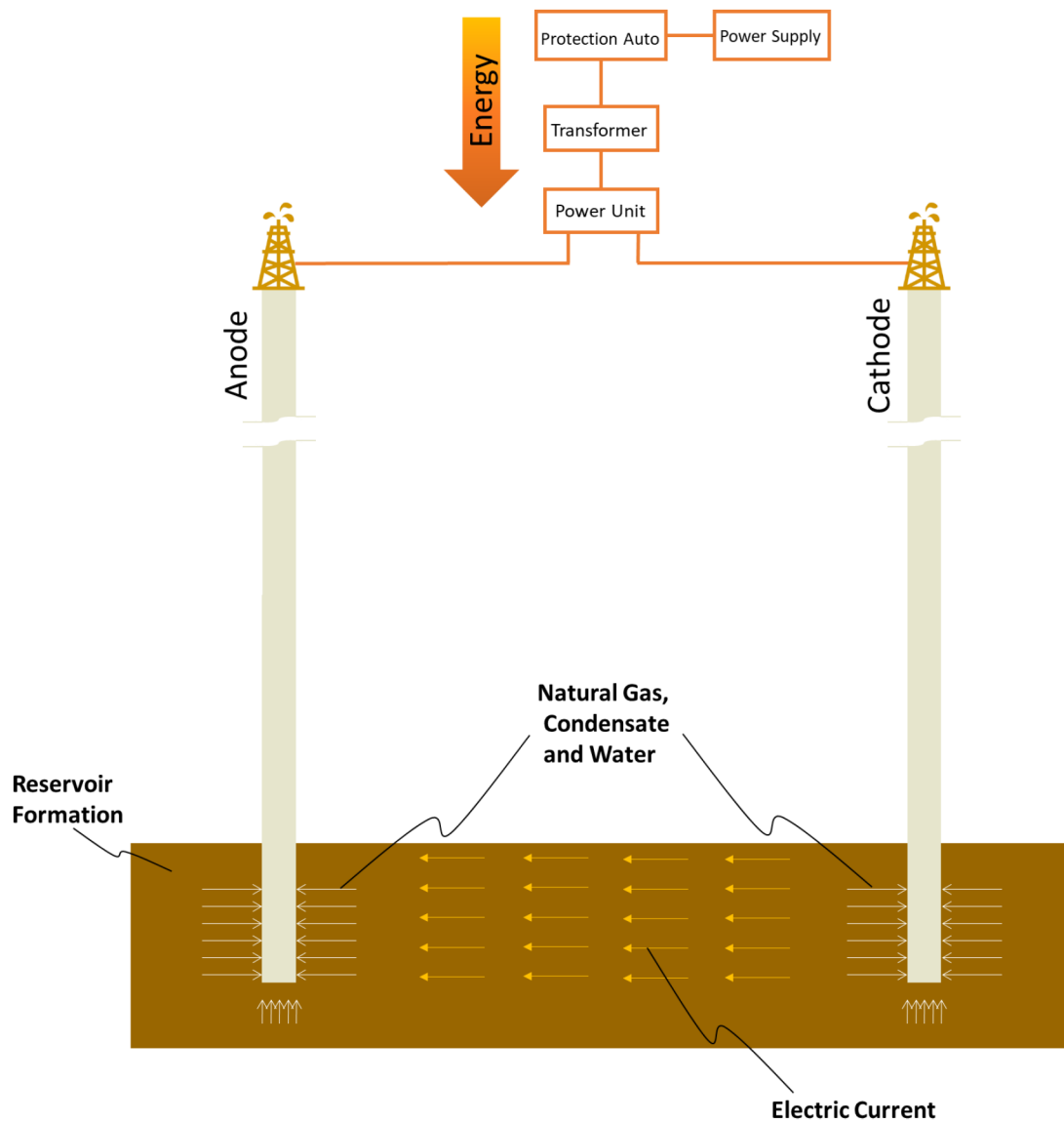


Figure 5—A schematic diagram of electrical arrangement which involve the placement of two electrodes between two dead gas vertical wells blocked by water. One well acting as source (cathode), whereas other well acting as sink (Anode), (Al-Juhani et al., 2024).

The sequence and steps involved in the proposed electrical treatment method (Figure 6) are as follows:

1. The treatment process will initially start by applying a power supply.
2. Electrical arrangement could involve the placement of two electrodes in between two spaced wells or within the same well, one acting as source and the other as a sink, wherein one of the electrodes is a cathode and the other is an anode and can cover a region of around 2-3 km. The electrical induced effects into the reservoir may vary according to the variation of the current density/voltage applied in between two spaced electrodes.
3. The current will be applied to the wellhead and the charge will propagate through the metallic casing all the way to the pay zone where the charge is delivered to subterranean reservoir to achieve a desired effect.
4. The local current pulses in the reservoir will enlarge the pore throats leading to some of the water to be released through porous media.
5. The local energy pulses will also cause partial electrolysis forming gas droplets from some released water.

6. The major proportion of water droplets released will be connected due to electrostatic coalescence for easy mobilization and removal. Figures 7 and 8 below provide the schematic illustrations of steps 4-6:

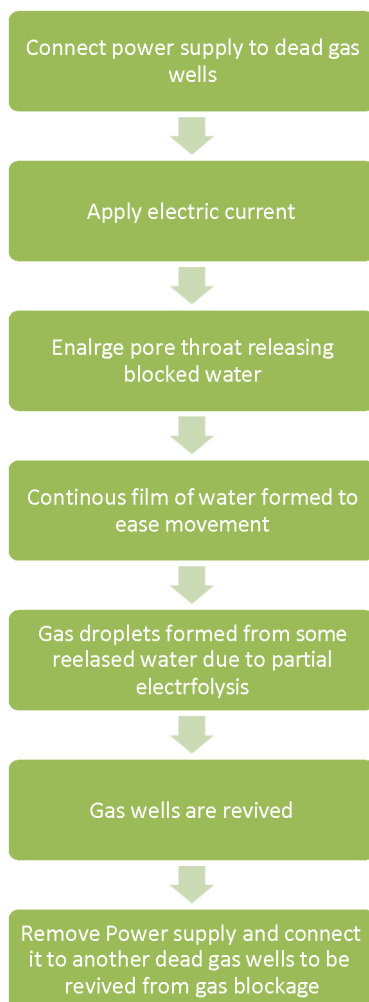


Figure 6—A workflow showing the summary of steps involved in the electrical treatment process from connecting power supply to applying electric current to revive dead gas wells until blockage is removed (Al-Juhani et al., 2024)

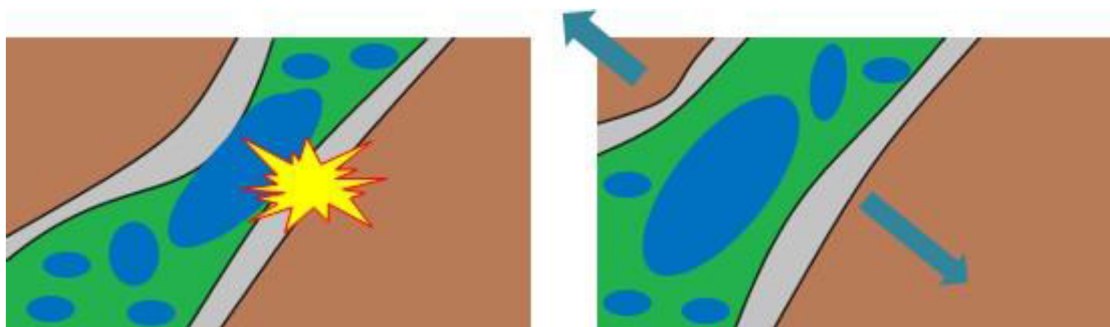


Figure 7—An illustration of increase and expansion in pore throat by applying an electric current. The left image represents before electric treatment; tight and small pore throat is formed due to strong attraction between positive poles of water molecules to surface and positive cations. The right image is a representation of increase in pore throat radius due to motion of water molecules, cations and anions. Due to electric current, partial electrolysis occurs releasing gas bubbles from water molecules (Al-Juhani et al., 2024)



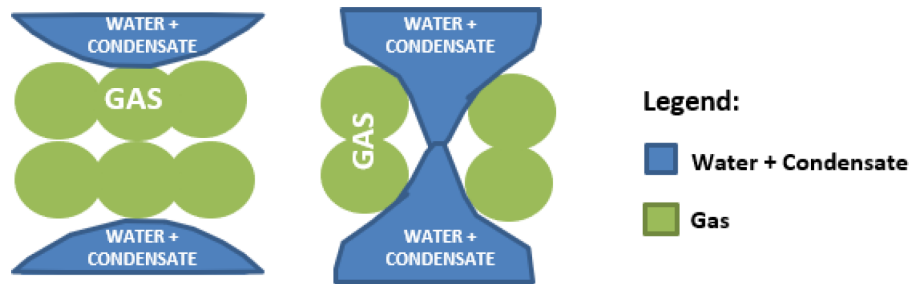


Figure 8—A representation of electric current application where the water/condensate droplets form a larger ganglia due to coalescence. This would form a continuous film of water/condensate to minimize surface energy and ease their movement (Al-Juhani et al., 2024)

The electrical treatment operation can take up to 30 hours, which can permanently remove the near wellbore water blockage due to effective release and mobilization of all the water clogging the pore throats. The treatment process will also simultaneously revive two dead gas wells as the current will flow between the two wells in the reservoir using formation water as the conductive medium.

When combining the two methods of electrical current and FSD for reviving aging gas wells (Figure 9), that are challenged with liquid loaded adverse situations, the result will yield production of the gas and the liquids at a much faster pace. Because the FSD extends the gas well operability range by increasing the gas velocity through a minor reduction in the cross-sectional flow area, and operating the well in the stable annular flow regime thereby stabilizing the annular film and preventing film reversal.

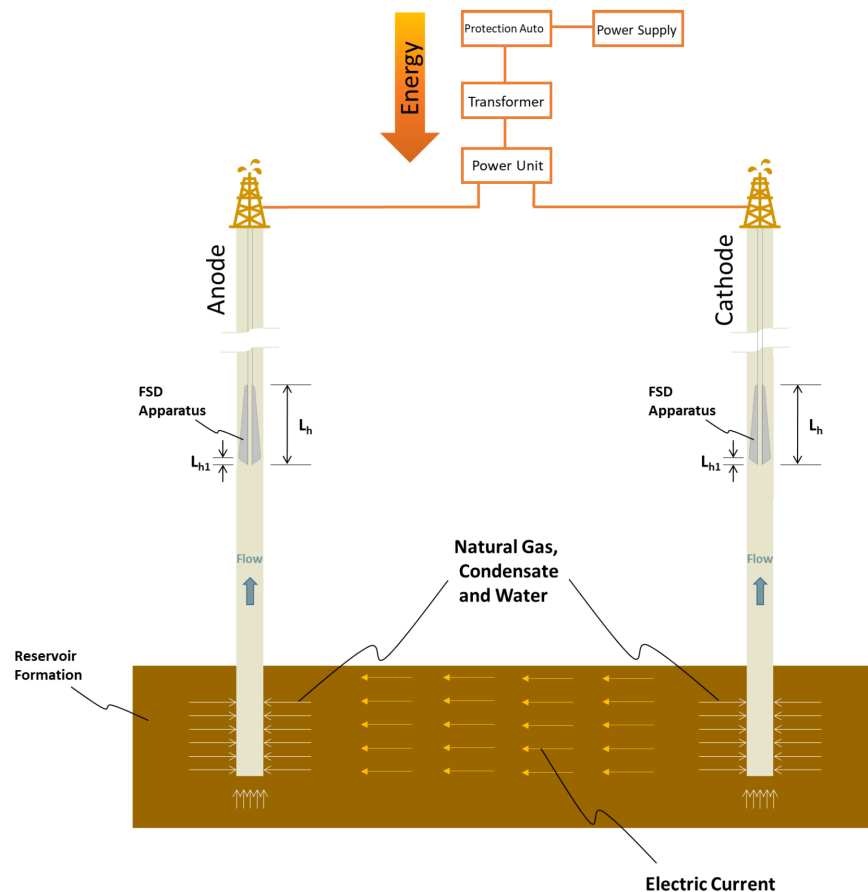


Figure 9—A schematic diagram of electrical arrangement, which involve the placement of two electrodes between two dead gas vertical wells blocked by water, combined with the flow stabilization device (FSD). One well acting as source (cathode), whereas other well acting as sink (Anode).



The gradual restriction of the inverse venturi geometry minimizes pressure drop due to the FSD and optimizes the pressure recovery downstream of the FSD. A benefit of the FSD is the ability to sustain higher flow rates due to the pressure recovery with the FSD geometry. The gradual restriction of the inverse venturi geometry minimizes pressure drop due to the FSD and optimizes the pressure recovery downstream of the FSD. A benefit of the FSD is the ability to sustain higher flow rates due to the pressure recovery with the FSD geometry.

## Technical Insights

Extensive Computational Fluid Dynamics (CFD) simulation work was conducted in order to prove the efficiency of the FSD concept and optimize the FSD design for the purpose of stabilizing the liquid film at the inner wall of the production column, in the range of flowrates and Gas Volume Fraction (GVF) of interest towards the practical application (Shariff et al. 2021). The results from the CFD analysis shown the FSD extends the gas well operability range by increasing the gas velocity through a minor reduction in the crosssectional flow area, and operating the well in the stable annular flow regime thereby stabilizing the annular film and preventing film reversal. The gradual restriction of the inverse venturi geometry minimizes pressure drop due to the FSD and optimizes the pressure recovery downstream of the FSD. A major benefit of the FSD is the ability to sustain higher flow rates due to the pressure recovery with the FSD geometry. The gradual restriction of the inverse venturi geometry minimizes pressure drop due to the FSD and optimizes the pressure recovery downstream of the FSD.

A growing number of gas wells around the world are challenged with liquid loading issues. These wells are kept shut-in or suspended when condensate/formation liquids cannot be lifted with the declining reservoir pressure and production tubing. Gas well life is typically about 20 years. Over that period reservoir pressure declines from original reservoir pressure (around 7,500 for certain reservoirs and 8,500 psi for other reservoirs) down to 3,000 psi or lower. Once reservoir pressures decline to around 3,000 psi or lower, the wells can no longer be produced against current flowline pressures without additional compression in addition to the increase in liquid influx and must be shut-in leaving valuable hydrocarbon in place. Oil and gas companies currently operates several 100's of gas production wells with a number of them having known production problems attributed to the liquid loading phenomenon. It is estimated that such number of problematic wells will also substantially increase in the coming years. Most gas wells drilled in the 1990's and early 2000's used a standardized approach resulting in a standard well completion with either 4-1/2" or 5-1/2" diameter production tubing. Therefore, the present technology primarily targets wells with the 4-1/2" and 5-1/2" diameter production tubing, but the present remedial intervention could potentially work for larger diameter production wells.

Good candidates for installing the FSD apparatus are gas wells of certain reservoir formations. These gas wells are 12,000 to 17,000 feet deep. The reservoir pressures range from 3,500 to 9,000 psi and the temperature varies from 250 to 350 °F. The method utilizes procedures for installing downhole equipment either on coiled tubing or utilizing a workover rig, depending on the operational capabilities of each unit. The installation configuration must be safe, cost-optimal, reproducible and reliable. Liquid loading of gas wells is a subject of high importance. Mitigating liquid loading translates to considerable improvement in the production of an aging gas well. In gas wells, natural gas, condensate and water are produced concurrently in the early stages of well life, the gas flow rate is high enough to lift liquid condensate all the way up to the ground level. Under these conditions, the flow regime is annular with gas flowing continuously in the core of the pipe and the liquid as entrained droplets flowing in the core surrounded by a liquid film on the wall. As the gas well matures, the produced gas flow rate and pressure decrease, reducing liquid lifting capability and initiating what is known as liquid loading. Liquid begins accumulating downhole and the flow pattern changes from annular to an intermittent and unsteady churning flow regime. The accumulated liquid hinders hydrocarbon production. The proposed remedies presented here extends the life of a producing well by

promoting a steady flow regime and transport of the gas and liquid phases; therefore, stabilizing the flow. The electrical treatment process can be effective to remove the water/condensate blockage in gas wells. The FSD when combined with the electrical treatment should prolong the life of these wells with much higher production rates of gas and condensates.

Currently, the most common conventional method for re-energizing a produced fluid flow upward a producing well is by increasing the flow rate in the existing production tubing by injecting a gas at high pressure at the appropriate height in the well. This technique is broadly named as "artificial lift" and is widely used to draw gas from wells with decreasing flowing pressure. Other conventional methods for pumping low pressure wells in the Oil & Gas industry include the use of a down-hole plunger or rod-type pump and the injection of chemicals to change the properties (density and viscosity) of the liquid phase and facilitate the gas production. Alternatively, a commercial solution for re-energizing gas wells that are choking due to liquid loading available to the gas production industry is known as "velocity string". The velocity string is simply a production tubing with smaller diameter than the initial well tubing which is installed in the well during a work over procedure to extend from the well bottom in the "pay zone" all the way up to the well head. Once the velocity string is installed, the annular space between the well tubing and the outside walls of the velocity string is shut and the well production is flowed through the velocity string. The production flow being produced through a smaller diameter tubing, the gas flowing speed is increased and the liquid can be entrained upwards the well, thus eliminating the liquid loading problem. More specifically, the current technique for mitigating liquid loading issues in gas wells is through the insertion of coiled tubing inside the 4-1/2" or 5-1/2" production tubing. This coiled tubing is also known as the "velocity string". This technique helps lift the gas at the cost of a very low production rate, typically 0.5 to 9 MMSCFD, as compared with the production rates for production size tubing. The 4-1/2" production liner produces 3.5 to 35 MMSCFD; the 5-1/2" production liner produces 5 to 45 MMSCFD under the same reservoir pressures.

## Summary and Conclusions

In this paper, the application of an integrated electrical treatment and FSD technology is proposed as a new solution to revive dead gas wells and sustain the gas production rates. Electric current is first applied either in a single well or both wells and charge will propagate through metallic casing. This would overcome the issue of water/condensate blockage to dead gas wells by reviving them and prolong their life. The proposed technical solution is to sustain a desired two-phase annular flow regime to transport liquid and gas upward inside the well by incorporating one or several Flow Stabilizing Devices (FSD). Thanks to its converging-diverging geometry, FSD increases the gas phase velocity while keeping the liquid phase transported as a film on the inner wall of the well. This avoids the liquid loading problem in the gas production well, thus extending its life, increasing its cumulative gas production and recovering the liquid condensate. The FSD configuration can be designed for the purpose of accelerating the flow at critical locations in the well where additional momentum is needed to sustain the upward liquid film flow.

Computational Fluid Dynamics work was conducted as a proof of concept to prove the efficiency and optimize the FSD design. More CFD work should be conducted to estimate the FSD performance to maintain an annular flow regime for lower gas production rates, and determine the minimum and maximum FSD vertical spacing, or height of influence, in the production column.

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